Trigonometry Equation

11. If $\theta \in [-2\pi, 2\pi]$, then the number of solutions of $2\sqrt{2}\cos^2\theta + \left(2-\sqrt{6}\right)\cos\theta - \sqrt{3} = 0$, is equal to :

यदि $\theta \in [-2\pi, 2\pi]$ है, तो समीकरण $2\sqrt{2}\cos^2\theta + \left(2-\sqrt{6}\right)\cos\theta - \sqrt{3} = 0$ के हलों की संख्या है :

(A) 6 (B) 10 (C) 12 (D) 8

Ans. D

Sol. $2\sqrt{2}\cos^2\theta + 2\cos\theta - \sqrt{6}\cos\theta - \sqrt{3} = 0$

$\left(2\cos\theta - \sqrt{3}\right)\left(\sqrt{2}\cos\theta + 1\right) = 0$

$\cos\theta = \frac{\sqrt{3}}{2}, -\frac{1}{\sqrt{2}}$

Number of solution = 8

12. The number of solutions of the equation $\left(4-\sqrt{3}\right)\sin x - 2\sqrt{3}\cos^2 x = -\frac{4}{1+\sqrt{3}}$, $x \in \left[-2\pi, \frac{5\pi}{2}\right]$ is :

समीकरण $\left(4-\sqrt{3}\right)\sin x - 2\sqrt{3}\cos^2 x = -\frac{4}{1+\sqrt{3}}$, $x \in \left[-2\pi, \frac{5\pi}{2}\right]$ के हलों की संख्या है :

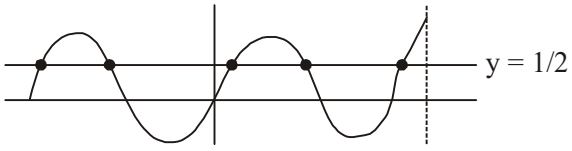
(A) 6 (B) 5 (C) 4 (D) 3

Ans. B

Sol. $\left(4-\sqrt{3}\right)\sin x - 2\sqrt{3}\left(1-\sin^2 x\right) = 2\left(1-\sqrt{3}\right)$

$\Rightarrow \left(2\sin x - 1\right)\left(\sqrt{3}\sin x + 2\right) = 0$

$\Rightarrow \sin x = \frac{1}{2}$



So number of solutions = 5

13. The number of solutions of the equation $\cos 2\theta \cos\frac{\theta}{2} + \cos\frac{5\theta}{2} = 2\cos^3\frac{5\theta}{2}$ in $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ is :

समीकरण $\cos 2\theta \cos\frac{\theta}{2}+\cos\frac{5\theta}{2}=2\cos^3\frac{5\theta}{2}$ के $\left[-\frac{\pi}{2},\frac{\pi}{2}\right]$ में हलों की संख्या है :

(A) 5 (B) 7 (C) 9 (D) 6

Ans. B

Sol. $\cos 2\theta\cdot\cos\frac{\theta}{2}=\cos\frac{5\theta}{2}\left(2\cos^2\frac{5\theta}{2}-1\right)$

$\Rightarrow 2\cos 2\theta\cdot\cos\frac{\theta}{2}=2\cos\frac{5\theta}{2}\cdot\cos 5\theta$

$\Rightarrow \cos\frac{5\theta}{2}+\cos\frac{3\theta}{2}=\cos\frac{15\theta}{2}+\cos\frac{5\theta}{2}$

$\Rightarrow \cos\frac{15\theta}{2}-\cos\frac{3\theta}{2}=0$

$\Rightarrow 2\sin\left(\frac{9\theta}{2}\right)\sin(3\theta)=0$

Either $\sin\left(\frac{9\theta}{2}\right)=0\Rightarrow\frac{9\theta}{2}=n\pi$

$\Rightarrow \theta=\frac{2n\pi}{9}\Rightarrow\theta=-\frac{4\pi}{9},-\frac{2\pi}{9},0,\frac{2\pi}{9},\frac{4\pi}{9}$

or $\sin(3\theta)=0$

$\Rightarrow 3\theta=n\pi$

$\Rightarrow \theta=\frac{n\pi}{3}$

$\Rightarrow \theta=-\frac{\pi}{3},0,\frac{\pi}{3}$

So, total solutions = 7

14. Let $A=\left\{x\in(0,\pi)-\left\{\frac{\pi}{2}\right\}:\log_{(2/\pi)}|\sin x|+\log_{(2/\pi)}|\cos x|=2\right\}$ and

$B=\left\{x\geq 0:\sqrt{x}\left(\sqrt{x}-4\right)-3\left|\sqrt{x}-2\right|+6=0\right\}$. Then $n(A\cup B)$ is equal to :

माना कि $A=\left\{x\in(0,\pi)-\left\{\frac{\pi}{2}\right\}:\log_{(2/\pi)}|\sin x|+\log_{(2/\pi)}|\cos x|=2\right\}$ और

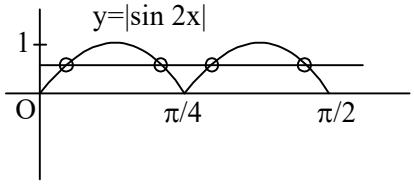
$B=\left\{x\geq 0:\sqrt{x}\left(\sqrt{x}-4\right)-3\left|\sqrt{x}-2\right|+6=0\right\}$ है। तब $n(A\cup B)$ बराबर है :

(A) 4 (B) 2 (C) 6 (D) 8

Ans. D

Sol. $A : |\sin x \cos x| = \left(\frac{2}{\pi}\right)^2$

$|\sin 2x| = \frac{8}{\pi^2}$



(4 solutions)

B : Put $\sqrt{x} - 2 = t$

$(t^2 - 4) - 3|t| + 6 = 0$

$|t|^2 - 3|t| + 2 = 0$

$|t| = 1, 2$ $\quad$ $x = 1, 9; 0, 16$ (4 solutions)

15. Suppose $\theta \in \left[0, \frac{\pi}{4}\right]$ is a solutions of $4\cos\theta - 3\sin\theta = 1$. Then $\cos\theta$ is equal to :

माना $4\cos\theta - 3\sin\theta = 1$ का एक हल $\theta \in \left[0, \frac{\pi}{4}\right]$ है। तो $\cos\theta$ बराबर है :

(A) $\frac{4}{\left(3\sqrt{6}+2\right)}$ $\quad$ (B) $\frac{4}{\left(3\sqrt{6}-2\right)}$ $\quad$ (C) $\frac{6-\sqrt{6}}{\left(3\sqrt{6}-2\right)}$ $\quad$ (D) $\frac{6+\sqrt{6}}{\left(3\sqrt{6}+2\right)}$

Ans. B

Sol. $4\cos\theta - 3\sin\theta = 1$

$4\cos\theta - 1 = 3\sin\theta$

$16\cos^2\theta + 1 - 8\cos\theta = 9(1 - \cos^2\theta)$

$\Rightarrow 25\cos^2\theta - 8\cos\theta - 8 = 0$

$\Rightarrow \cos\theta = \frac{8 \pm \sqrt{64 + 4 \times 25 \times 8}}{2.25}$

$= \frac{8 \pm 4\sqrt{4 + 50}}{2.25}$

$= \frac{4 \pm 2\sqrt{54}}{25}$

As $\theta \in \left[0, \frac{\pi}{4}\right]$

$\Rightarrow \cos\theta = \frac{4+6\sqrt{6}}{25} = \frac{4}{3\sqrt{6}-2}$

16. If $2\tan^2\theta - 5\sec\theta = 1$ has exactly 7 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, for the least value of $n \in N$, then $\sum_{k=1}^{n} \frac{k}{2^k}$ is equal to :

जिस $n \in N$ के न्यूनतम मान के लिए अंतराल $\left[0, \frac{n\pi}{2}\right]$ में $2\tan^2\theta - 5\sec\theta = 1$ के ठीक 7 हल हैं, उस n के लिए $\sum_{k=1}^{n} \frac{k}{2^k}$ बराबर है :s

(A) $\frac{1}{2^{14}}\left(2^{15}-15\right)$ (B) $1-\frac{15}{2^{13}}$ (C) $\frac{1}{2^{13}}\left(2^{14}-15\right)$ (D) $\frac{1}{2^{15}}\left(2^{14}-14\right)$

Ans. C

Sol. $2\tan^2\theta - 5\sec\theta - 1 = 0$

$\Rightarrow 2\sec^2\theta - 5\sec\theta - 3 = 0$

$\Rightarrow (2\sec\theta + 1)(\sec\theta - 3) = 0$

$\Rightarrow \sec\theta = -\frac{1}{2}, 3$

$\Rightarrow \cos\theta = -2, \frac{1}{3}$

$\Rightarrow \cos\theta = \frac{1}{3}$

For 7 solutions n = 13

So, $\sum_{k=1}^{13} \frac{k}{2^k} = S$ (say)

$S = \frac{1}{2} + \frac{2}{2^2} + \frac{3}{2^3} + \ldots + \frac{13}{2^{13}}$

$\frac{1}{2}S = \frac{1}{2^2} + \frac{1}{2^3} + \ldots + \frac{12}{2^{13}} + \frac{13}{2^{14}}$

$$\Rightarrow \frac{S}{2}=\frac{1}{2}\cdot\frac{1-\frac{1}{2^{13}}}{1-\frac{1}{2}}-\frac{13}{2^{14}}$$

$$\Rightarrow S=2\cdot\left(\frac{2^{13}-1}{2^{13}}\right)-\frac{13}{2^{13}}$$

17. If $\alpha, -\frac{\pi}{2}<\alpha<\frac{\pi}{2}$ is the solution of $4\cos\theta+5\sin\theta=1$, then the value of $\tan\alpha$ is :

यदि समीकरण $4\cos\theta + 5\sin\theta = 1$ का हल $\alpha, -\frac{\pi}{2}<\alpha<\frac{\pi}{2}$ है, तो $\tan\alpha$ का मान है :

(A) $\frac{10-\sqrt{10}}{12}$ (B) $\frac{\sqrt{10}-10}{6}$ (C) $\frac{10-\sqrt{10}}{6}$ (D) $\frac{\sqrt{10}-10}{12}$

Ans. D

Sol. $4 + 5\tan\theta = \sec\theta$

Squaring : $24\tan^2\theta + 40\tan\theta + 15 = 0$

$$\tan\theta=\frac{-10\pm\sqrt{10}}{12}$$

and $\tan\theta=-\left(\frac{10+\sqrt{10}}{12}\right)$ is Rejected

18. The sum of the solutions $x \in R$ of the equation $\frac{3\cos 2x+\cos^3 2x}{\cos^6 x-\sin^6 x}=x^3-x^2+6$ is :

समीकरण $\frac{3\cos 2x+\cos^3 2x}{\cos^6 x-\sin^6 x}=x^3-x^2+6$ के हलों $x \in R$ का योगफल है :

(A) 1 (B) 0 (C) –1 (D) 3

Ans. A

Sol. Given $\frac{3\cos 2x+\cos^3 2x}{\cos^6 x-\sin^6 x}=x^3-x^2+6$

$$\Rightarrow \frac{\cos 2x\left(3+\cos^2 2x\right)}{\left(\cos^2 x-\sin^2 x\right)\left[\sin^4 x+\cos^4 x+\sin^2 x\cos^2 x\right]}$$

$$= x^3-x^2+6$$

$$\Rightarrow \frac{3+\cos^2 2x}{1-\sin^2 x\cos^2 x} = x^3 - x^2 + 6$$

$$\Rightarrow 4\left(\frac{3+\cos^2 2x}{4-\sin^2 2x}\right) = x^3 - x^2 + 6$$

$$\Rightarrow x^3 - x^2 + 2 = 0$$

$$\Rightarrow (x+1)\left(x^2 - 2x + 2\right) = 0$$

$$\Rightarrow (x+1)\left((x-1)^2 + 1\right) = 0$$

$\therefore$ Sum of solutions $= -1$

19. If $2\sin^3 x + \sin 2x \cos x + 4\sin x - 4 = 0$ has exactly 3 solutions in the interval $\left[0, \frac{n\pi}{2}\right]$, $n \in N$, then the roots of the equation $x^2 + nx + (n-3) = 0$ belongs to :

यदि $2\sin^3 x + \sin 2x \cos x + 4\sin x - 4 = 0$ के अंतराल $\left[0, \frac{n\pi}{2}\right]$, $n \in N$ में ठीक 3 हल हैं, तो समीकरण $x^2 + nx + (n-3) = 0$ के मूल किस में हैं?

(A) $(-\infty, 0)$ (B) $\left(-\frac{\sqrt{17}}{2}, \frac{\sqrt{17}}{2}\right)$ (C) $(0, \infty)$ (D) Z

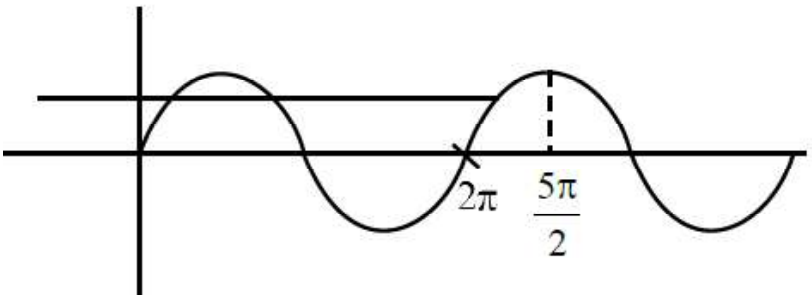
Ans. A

Sol. $2\sin^3 x + 2\sin x \cos^2 x + 4\sin x - 4 = 0$

$\Rightarrow 2\sin^3 x + 2\sin x (1 - \sin^2 x) + 4\sin x - 4 = 0$

$\Rightarrow 2\sin^3 x + 6\sin x - 2\sin^3 x - 4 = 0$

$\sin x = \frac{2}{3}$



$\left[0, \frac{5\pi}{2}\right]$

$x^2 + 5x + 2 = 0$

$\alpha, \beta = \frac{-5 \pm \sqrt{17}}{2}$ both negative, hence $(-\infty, 0)$

20. The number of solutions of the equation $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0; x \in [-2\pi, 2\pi]$ is :

समीकरण $4\sin^2 x - 4\cos^3 x + 9 - 4\cos x = 0; x \in [-2\pi, 2\pi]$ के हलों की संख्या है :

(A) 0 (B) 3 (C) 1 (D) 2

Ans. A

Sol. $4 - 4\cos^2 x - 4\cos^3 x + 9 - 4\cos x = 0$

$4\cos^3 x + 4\cos^2 x + 4\cos x = 13$

$\because$ LHS ≤ 12

So equation will have no solution.

21. Let S = $\{\sin^2 2\theta : (\sin^4\theta + \cos^4\theta)x^2 + (\sin 2\theta)x + (\sin^6\theta + \cos^6\theta) = 0$ has real roots$\}$. If α and β be the smallest and largest elements of the set S, respectively, then $3((\alpha - 2)^2 + (\beta - 1)^2)$ equals _____.

माना S = $\{\sin^2 2\theta : (\sin^4\theta + \cos^4\theta)x^2 + (\sin 2\theta)x + (\sin^6\theta + \cos^6\theta) = 0$ के वास्तविक मूल हैं$\}$ हैं। यदि α तथा β समुच्चय S के क्रमशः सबसे छोटा तथा सबसे बड़ा अवयव हैं, तो $3((\alpha - 2)^2 + (\beta - 1)^2)$ बराबर है _____

Ans. 4

Sol. $D \ge 0$

$\Rightarrow \sin^2 2\theta - 4\left(\sin^4\theta + \cos^4\theta\right)\left(\sin^6\theta + \cos^6\theta\right) \ge 0$

$\Rightarrow t - 4\left(1 - \frac{t}{2}\right)\left(1 - \frac{3t}{4}\right) \ge 0$ (Let $\sin^2 2\theta = t$)

$\Rightarrow 3t^2 - 12t + 8 \le 0$

$\Rightarrow t \in \left[2 - \frac{2}{\sqrt{3}}, 2 + \frac{2}{\sqrt{3}}\right]$ _____(1)

But $t = \sin^2 2\theta \in [0, 1]$ _____(2)

(A) $\cap$ (2) $t \in \left[2 - \frac{2}{\sqrt{3}}, 1\right]$

$\alpha = 2 - \frac{2}{\sqrt{3}}$, $\beta = 1$

$3((\alpha - 2)^2 + (\beta - 1)^2) = 4$

22. The number of solutions of $\sin^2 x + (2 + 2x - x^2)\sin x - 3(x - 1)^2 = 0$, where $-\pi \le x \le \pi$, is _____.

$\sin^2 x + (2 + 2x - x^2) \sin x - 3 (x - 1)^2 = 0$, जहाँ $-\pi \le x \le \pi$ के हलों की संख्या है _____

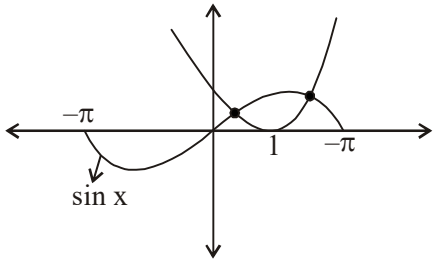
Ans. 2

Sol. $\sin^2 x + (3 - (x - 1)^2) \sin x - 3(x - 1)^2 = 0$

$\sin^2 x + 3 \sin x - (x - 1)^2 \sin x - 3(x - 1)^2 = 0$

$\sin x (\sin x + 3) - (x - 1)^2) [\sin x + 3] = 0$

$\underbrace{(\sin x + 3)}_{\substack{\sin x \neq -3 \\ \downarrow \\ \text{not possible}}}$ $\qquad$ $\underbrace{\left(\sin x - (x-1)^2\right)}_{\sin x = (x-1)^2} = 0$



There are two intersections between this graph.

So, number of solution will be 2.

23. Let the set of all $a \in R$ such that the equation $\cos 2x + a \sin x = 2a - 7$ has a solution be [p, q] and $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$, then pqr is equal to _____.

माना $a \in R$ के सभी मानों, जिनके लिए समीकरण $\cos 2x + a \sin x = 2a - 7$ का एक हल है, का समुच्चय [p, q] है तथा $r = \tan 9^\circ - \tan 27^\circ - \frac{1}{\cot 63^\circ} + \tan 81^\circ$ है। तो pqr बराबर है _____।

Ans. 48

Sol. $\cos 2x + a \cdot \sin x = 2a - 7$

$a(\sin x - 2) = 2(\sin x - 2)(\sin x + 2)$

$\sin x = 2, a = 2(\sin x + 2)$

$\Rightarrow a \in [2, 6]$

$p = 2 \quad q = 6$

$r = \tan 9^\circ + \cot 9^\circ - \tan 27 - \cot 27$

$r = \frac{1}{\sin 9 \cdot \cos 9} - \frac{1}{\sin 27 \cdot \cos 27}$

$$= 2\left[\frac{4}{\sqrt{5}-1} - \frac{4}{\sqrt{5}+1}\right]$$

$r = 4$

$p \cdot q \cdot r = 2 \times 6 \times 4 = 48$